

Towards an Optimal Feature Selection Method for AI-Based DDoS Detection System

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Abstract—Cyber-attacks are increasing rapidly, so developing effective intrusion detection and prevention tools for a secure and safer cyberspace is crucial. DDoS (Distributed Denial of Services) is one of the most well-known digital threats, endangering any cyber-physical system. DDoS prevents the host from serving the legitimate traffic by overflowing the host node with unwanted service requests. Nowadays, machine learning-based IDS (Intrusion Detection System) uses different Feature Selection (FS) methods to extract a feature subset from a large dataset to increase the model performance and decrease the training time. In this research work, we used the UNSW-NB15 dataset [1] to conduct a comprehensive analysis for evaluating the performance of different FS techniques in DDoS attack classification using both Machine Learning (ML) and Deep Learning (DL) models. Furthermore, an Ensemble Feature Selection (EN-FS) technique called Majority Voting (MV) has been implemented to combine the individual FS method's output to extract an optimal feature set. Our ensemble feature selection approach significantly reduces the features from 42 to 15, which is 64% less than the original features. Lastly, an extensive experiment has been performed to estimate and compare the performance of individual, ensemble, and original feature set in both ML and DL-based DDoS detection systems. According to our analysis, the ensemble feature set-based classification model exhibits higher accuracy, lower False Positive Rate (FPR), and better execution time than the other individual feature set-based models.

Index Terms—DDoS, cyber-attack, ensemble feature selection, majority voting, intrusion detection model, machine learning, deep learning

I. INTRODUCTION

Internet Service Providers (ISPs) all around the globe are experiencing a significant increase in network attacks in recent years. They are constantly seeking effective and intelligent solutions to keep the Internet safe and secure from the cyber attackers. Attackers try to exploit the vulnerable location to access and utilize information to support themselves or obstruct institutions or businesses. As a result, cyber-security has become a primary priority for the technological world.

DDoS (Distributed Denial of Service) is one of the most prevalent cyber-attack techniques. Under DDoS attack the network is overburdened with messages, making it unable to process genuine requests from customers or other network components. Machine Learning (ML)-based Intrusion Detection Systems (IDSs) are being used to defend against such attacks. It will improve the precision, efficiency, and resilience of networked information infrastructures compared

to older systems. In amalgamation with IDS, Feature Selection (FS) methods eliminate redundant data from the dataset and improve generalization capabilities as it needs fewer attributes and takes less time to analyze. According to Brownlee et al [2], FS improves model interpretability while simultaneously dropping the system's possibilities of overfitting. Moreover, feature selection aids data preparation and reduces data dimensionality [3].

This research proposes an ensemble feature selection technique to select an optimal feature set for both Machine Learning (ML) and Deep Learning (DL) models. There are three major types of Feature Selection (FS) methods as Filter-based, Wrapper-based, and Embedded explained in detail by Farhan et al [4]. Filter-based feature selection methods identify those features which have a strong statistical correlation with the target variable. This approach is comparatively faster than the other feature selection algorithms with low computational complexity. The wrapper-based feature selection method estimates the accuracy of a particular classification algorithm to identify a feature set that substantially makes a learning difference. This approach continuously updates an initial feature subset by adding or removing the feature based on the learning model's inferences. The embedded feature selection method combines both filter-based and wrapper-based techniques, and this method uses its feature selection algorithm to identify the best feature subset from the dataset. Géron et al. [5]. highlights that the ensemble approach employs a variety of strategies, including (i) Majority Voting, (ii) Bagging and Posting, (iii) Boosting (iv) Random Forest, and (v) Stacking.

Our proposed work selects fifteen different FS methods from three main categories to choose a feature subset. In this paper, we focused on exploring a combination of Feature Selection methods from all three categories, which were not investigated in the previous works. This approach is used to find the optimal feature set. Each feature subset extracted from the individual FS method shows varying classification accuracy in ML and DL models. So, we proposed an ensemble feature selection technique that combines the result from individual FS methods to generate an optimum feature subset.

This paper organized as follows. The sections follow as: Section II describes the literature review of current intrusion detection systems and related work in machine learning and deep learning. Furthermore, Section III presents the proposed

methodology to architect the ensemble feature selection-based IDS. The expected performance results of the different FS methods in both ML and DL models are presented in Section IV. Lastly, Section V concludes our paper and outlines the ideas for future research actions.

II. LITERATURE REVIEW

Dash and Liu [6] extract features from the dataset by using several feature selection techniques. After that, they tried to apply several algorithms such as Naive Bayes, Support Vector Machine, Decision Tree, etc. were applied on it to find the possible and best result for an Intrusion Detection System, which can help in reducing the cyber-attacks as well as can predict whether the system is malicious or not. In addition, supervised and unsupervised ensemble frameworks were also used to detect DDoS attacks with better accuracy. Chandrashekhara and Sahin [7] provided an in-depth study on FS based on both supervised learning and unsupervised learning. Indication of significant improvement in outcomes utilizing the FS strategy on calculations execution is shown. Sheikhpour, Razieh, et al. [8] additionally focused on semi-supervised FS and introduced two scientific classifications. Khalid, Samina et al. [9] scrutinized the dimensionality reduction techniques to evaluate the efficacy and shortcomings of these techniques. They have projected a survey of FS and extraction in this research work to improve data analysis. Luis, Molina, et al. [10] concocted a measure to get the level to estimate under controlled examinations on the same data set between FSA arrangement and factual arrangement. For assessment measures, they considered three features for the informational index. Their study shows that these selective criteria of FSA show firm dependence on data analysis. Adams and Beling [11] surveyed Gaussian Mixture Models (GMM) and Hidden Markov Models (HMM). They proposed that methods developed for one of these algorithms can be adapted for another and vice versa. In another research work, Das, Saikat et al. [12] proposed an ensemble framework for feature selection by introducing majority voting, which significantly increases the accuracy for detecting DDoS attacks.

Several limitations in the previous works, such as a limited number of FS algorithms used, lack of variation in FS method type, and the experiments limited to machine learning models, motivated us to propose an improved IDS model addressing the shortcomings mentioned earlier. We have chosen a combination of FS methods from filter-based, wrapper-based, and embedded types in our work and ensemble the individual FS results to identify a best-performing feature subset. The ensemble feature set outperforms the unique feature set in machine learning and deep learning models, which was not explored in current literature.

III. METHODOLOGY

A. Dataset

We have used the UNSW-NB15 dataset [1] created by the Australian Centre for Cyber Security (ACCS) in the Cyber Range Lab. We have explicitly focused on DDoS attacks in this

dataset. All other network flows have been removed from both the training and testing dataset for our experiment. There is a total of 112001 training instances in our dataset consisting of 56001 malicious traffics and 56000 benign traffics. The testing dataset contains 69996 instances divided into a malicious and benign set of equal size 34998 traffics.

B. Data Preprocessing

Data preprocessing is the very first crucial step in a machine learning solution. The ultimate performance of the designed machine learning model is highly dependent on the quality of the training data. Furthermore, the data processing techniques help us modify our data to be trainable, remove unexpected values such as infinity or null values, eliminate the unrealistic combination of data, etc. The UNSW-NB 15 dataset consists of both categorical and numerical values. In our experiment, the Label Encoder function from the sklearn library has been used, which transforms a list of categorical values into a list of corresponding numerical values. In addition, the missing, null, and infinity values are dropped from the dataset. Finally, a data normalization or features scaling technique is applied to our dataset, converting all input data to a standard scale. Another method transforms each feature to a given range (e.g., 0 to 1) by Min-Max Scaling, and this technique is used in our experiment.

C. Ensemble Feature Selection

There are three major types of FS methods as Filter-based FS method, Wrapper-based FS method, and Embedded FS method. In our experiment, we have selected a total of fifteen FS algorithms combining all different types. A total of seven filter-based FS method has been used, and they are Pearson's correlation (PEARSON), ANOVA, Chi-Square (CHI2), SelectFDR (SFDR), SelectFPR (SFPR), Select Percentile (SPERCENT), and Variance Threshold (VTSLED). In addition, mutual Information (MUTINFO) and Recursive Feature Elimination (RFE) methods are chosen from wrapper-based methods. Finally, we have selected LASSO Regression (LASSO), Logistic Regression (LRL1), Random Forests (RF), Light GBM (LGBM), Extra Tree (EXTREES), and Linear regression (FREGEX) as embedded FS methods. Each feature selection methods independently select a distinct feature set that suits the target classification better. Later, the ensemble technique called Majority Voting (MV) combines the features from all 15 methods. This technique selected those features voted by more than 50% of the feature selection algorithms (e.g., we have chosen a feature for ensemble set if it is selected by at least eight or more algorithms out of 15).

D. Software and Hardware Preliminaries

We have used Python, and the machine learning library scikit-learn [13], keras [14], TensorFlow to conduct the experiments in computers with the configuration of Intel (R) i3-8130U CPU@2.20GHz, 8GB memory, and 64-bit Windows operating system.

TABLE I: Extracted Features Using Ensemble Feature Selection Method

Feature	PEARSON	MUTINFO	SPERCENT	CHI2	ANOVA	FREGEX	SFPR	SFDR	LRL1	LASSO	RF	EXTREES	LGBM	RFE	VTSLD	Count
dttl	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	14
proto	1	0	1	1	1	1	1	1	1	1	1	1	0	1	1	13
dload	1	0	1	1	1	1	1	1	1	0	1	0	0	1	1	11
swin	1	0	0	1	1	1	1	1	1	1	0	1	0	1	1	11
sttl	1	1	1	1	1	1	1	1	0	0	1	1	0	0	1	11
ct_state_ttl	1	1	1	1	1	1	0	0	1	1	1	1	0	1	0	11
ct_srv_dst	1	0	0	1	1	1	0	0	1	1	1	1	1	1	0	10
rate	1	1	1	1	1	1	1	1	0	0	1	0	0	0	1	10
sload	1	1	0	1	1	1	1	1	0	0	1	0	1	0	1	10
sinpkt	1	0	1	1	1	1	1	1	0	0	0	0	1	0	1	9
dmean	1	0	1	1	1	1	0	0	1	0	1	0	1	1	0	9
service	1	0	0	1	1	1	1	1	1	0	0	0	0	0	1	8
ct_srv_src	1	0	0	1	1	1	0	0	0	1	1	1	0	1	0	8
is_sm																
ips_ports	1	0	1	1	1	1	0	0	1	1	0	0	0	1	0	8
sbytes	0	1	0	0	0	0	1	1	0	1	0	1	1	1	1	8

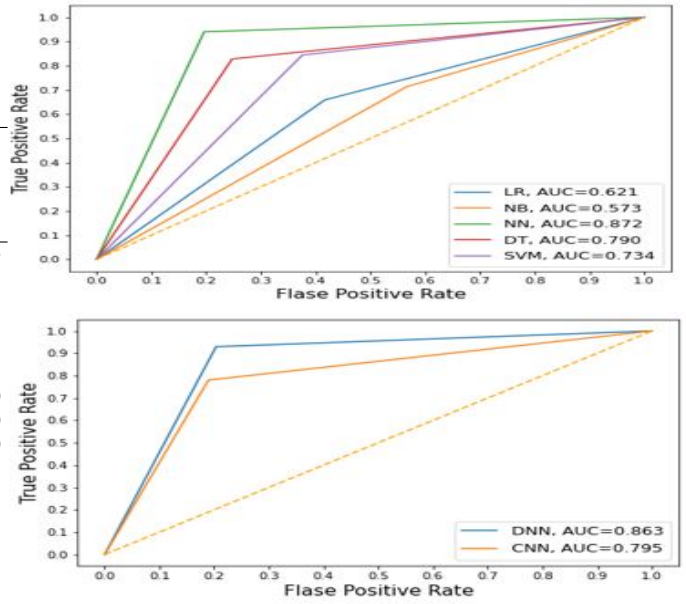


Fig. 1: Performance analysis

TABLE II: Best Performing Machine Learning Model for Particular Feature Selection Method

Feature Set	Classifier	Acc.	F1	Prec.	Rec.	FPR	Time(S)
PEARSON	NN	0.857	0.870	0.857	0.855	0.238	185.955
MUTINFO	NN	0.810	0.853	0.810	0.804	0.365	253.412
SPERCENT	DT	0.810	0.828	0.810	0.807	0.307	0.480
CHI2	NN	0.871	0.880	0.872	0.871	0.203	272.744
ANOVA	DT	0.851	0.865	0.851	0.850	0.245	0.975
FREGEX	DT	0.852	0.865	0.852	0.851	0.242	0.866
SFPR	NN	0.831	0.840	0.831	0.829	0.251	209.531
SFDR	NN	0.831	0.840	0.831	0.829	0.251	194.477
LRL1	NN	0.808	0.853	0.808	0.801	0.372	88.784
LASSO	NN	0.718	0.725	0.718	0.716	0.372	136.589
RF	NN	0.803	0.803	0.803	0.803	0.217	228.369
EXTREES	NN	0.807	0.810	0.807	0.806	0.247	250.997
LGBM	NN	0.684	0.685	0.684	0.684	0.279	342.910
RFE	LR	0.732	0.748	0.732	0.728	0.395	1.131
VTSLD	NN	0.806	0.856	0.806	0.799	0.381	188.540
ALL	NN	0.857	0.876	0.857	0.855	0.255	311.759
ENSEMBLE	NN	0.872	0.879	0.872	0.871	0.196	159.632

TABLE III: Best Performing Deep Learning Model for Particular Feature Selection Method

Feature Set	Classifier	Acc.	F1	Prec.	Rec.	FPR	Time(S)
PEARSON	CNN	0.854	0.866	0.799	0.945	0.237	165.796
MUTINFO	CNN	0.813	0.841	0.731	0.989	0.363	202.550
SPERCENT	DNN	0.804	0.834	0.723	0.987	0.378	142.429
CHI2	CNN	0.855	0.865	0.808	0.932	0.221	169.760
ANOVA	CNN	0.847	0.860	0.792	0.942	0.247	168.991
FREGEX	CNN	0.849	0.862	0.795	0.940	0.242	169.537
SFPR	CNN	0.807	0.837	0.723	0.994	0.380	169.228
SFDR	DNN	0.814	0.842	0.732	0.989	0.362	102.994
LRL1	DNN	0.828	0.849	0.756	0.967	0.311	103.038
LASSO	DNN	0.734	0.760	0.692	0.843	0.375	103.146
RF	CNN	0.734	0.716	0.768	0.671	0.202	163.063
EXTREES	DNN	0.815	0.822	0.793	0.854	0.223	104.458
LGBM	DNN	0.757	0.757	0.758	0.755	0.241	104.922
RFE	CNN	0.725	0.737	0.707	0.770	0.320	171.498
VTSLD	CNN	0.811	0.840	0.729	0.992	0.369	172.253
ALL	CNN	0.843	0.854	0.797	0.921	0.235	172.403
ENSEMBLE	DNN	0.863	0.871	0.820	0.929	0.204	142.672

TABLE IV: Comparisons with other methods

Research	EnFS	ML	DL	Accuracy	Time
[15]	N/A	Yes	No	86.01%	N/A
[16]	N/A	No	Yes	99.50%	N/A
[17]	N/A	Yes	No	90.85%	N/A
Our work	MV	Yes	Yes	80.4%, 87.7%	159.632

E. Evaluation Metrics

We have used six different evaluation metrics, which are: Accuracy (Acc.), Precision (Prec.), Recall (Rec.), F-1 Score (F1), False Positive Rate (FPR), and total execution time is second (Time(S)) to estimate the performance of our classification models. Besides, we have used another evaluation metric called Area Under the Curve- Receiver Operating Characteristics (AUC-ROC) curve to measure the performance of our classification model. ROC is a probability curve, and AUC represents the degree or measure of separability, measured based on the False Positive Rate (FPR) and True Positive Rate (TPR). The ROC curve is plotted with TPR in the y-axis vs. FPR in the x-axis and classification; the higher the ROC, the better the model is at distinguishing the classes.

IV. RESULTS AND DISCUSSION

We have analyzed and described the experimental results here to evaluate the performance of our proposed solution. Fifteen feature selection methods from three main categories: filter-based, wrapper-based, and embedded, have been applied to extract an individual set of best features. In addition, a grid search technique has been used to fine-tune the hyper-parameters for the FS methods. Table I represents a list of features selected by the ensembling technique called Majority Voting (MV). In MV, we have considered a feature more critical when selected by more than half of the candidate FS methods. Therefore, the count column in Table I represents

the total number of votes for an individual feature. The value 1 represents the selection of a feature by a particular feature selection method, while 0 indicates the rejection. So, according to our experimental result, we selected 15 out of 42 original features from the UNSWNB-15 dataset, which reduced the original feature set by more than 64%. Five different machine learning models have been trained on the feature sets selected by individual FS methods, the set of original features without any feature engineering, and the ensemble feature set selected by the ensemble FS technique. A total of 85 (17 feature sets used to train five individual model) machine learning models have been trained on our UNSWNB-15 dataset. Table II describes the performance metrics of the best-performing model for each particular feature set. We have summarized the model performance in terms of Accuracy (Acc.), F1-Score (F1), Precision (Prec.), Recall (Rec.), False Positive Rate (FPR), and total execution time in second (Time(S)). Among all the best performing models, the ensemble feature-based neural network shows the highest accuracy (87.2%) and lowest FPR (False Positive Rate) (19.6%). Also, the total execution time (159.632 seconds) is significantly lower by more than 41% execution time (272.744 seconds) of the second-best performing. Overall, the ensemble feature set-based machine learning model exhibits a comparatively higher accuracy, lower FPR, and better execution time, indicating the ensemble technique's advantages in extracting an optimal feature set. Since the ensemble feature set outperforms the other FS methods, the ROC curve for all machine learning and deep learning models based on the ensemble feature set is shown in Fig. 1. We also justify the best performance of the ensemble feature set using deep learning models. Two state-of-the-art deep learning models, such as DNN and CNN, have been implemented in our experiment to show a performance improvement of ensemble technique. Table III shows the best performing deep learning model's performance for the individual feature set. We have identified the best accuracy of 86.3% for ensemble feature-based DNN model with the lowest FPR of 20.4%. The execution time (142.672 sec.) is also lower by more than 13% of the second-best performing model's execution time (165.796 sec.). Our experimental result shows a performance improvement in DDoS detection using the ensemble feature selection technique. Both machine learning and deep learning models exhibit higher accuracy, lower FPR, and better execution time. Further, A comparative performance analysis between our and existing work is summarized in Table IV.

V. CONCLUSION

Our research explains the importance of selecting an optimal feature set to classify DDoS attacks using machine learning and deep learning algorithms. We have done a comprehensive comparative analysis among fifteen individual feature sets, ensemble feature sets, and original feature set to identify an optimal set of features. In terms of the ensemble FS method, we have used the Majority Voting technique that selects features identified as more relevant by more than 50%

FS methods. The extracted feature set has been used to train five ML algorithms and two DL models to evaluate the performance. After analyzing the result, we concluded that our EN-FS technique outperforms the individual FS method in both ML and DL models. In future work, we would like to improve the accuracy of our proposed machine learning models by ensembling the individual model's result. Furthermore, we will apply this technique to other datasets to further explore and analyze the effectiveness of our methodology.

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